UCQ 2025 EDUCATION SERIES LECTURE NOTES

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1. (Ryan) Math of Quantum Mechanics

Quantum States. In classical mechanics, we're used to particles being in certain states. For example a particle could be "to the left" or "to the right". We denote states using *kets*

$$|L\rangle : \text{particle is to the left}$$
$$|R\rangle : \text{particle is to the right}$$

What sets quantum mechanics apart is that particles can be in a *superposition*, essentially in multiple states at the same time! This means we have to be able to do two things to our states

- Add them together $|L\rangle + |R\rangle$ to represent superpositions
- Scale a state $c\,|L\rangle$ to represent probability in some sense (that we'll define later)

Together with some boring consistency conditions, this forms a mathematical structure called a *vector space*.

We also want a notion of size. This is measured by an *inner product* $\langle \cdot, \cdot \rangle$, analogous to the dot product. This way, we can measure the size of a state

$$|v|^2 = v \cdot v = v^T v \qquad \Longleftrightarrow \qquad |\,|\psi\rangle\,|^2 = \langle \psi, \psi \rangle$$

Together with more boring consistency conditions, this forms a mathematical structure called a *Hilbert space*.

Key Idea 1.1. The set of all quantum states forms a Hilbert space.

In our system, a general quantum state is given by any combination of possible states.

$$|\psi\rangle = a\,|L\rangle + b\,|R\rangle$$

For reasons we'll explore later, a and b are allowed to take on complex values. We can also construct the *dual* of a quantum state with a *bra*.

$$\langle\psi| = |\psi\rangle^\dagger$$

where the dagger is the *Hermitian conjugate* (transpose + complex conjugate)

$$|\psi\rangle^\dagger = (|\psi\rangle^T)^*$$

The complex conjugate is necessary so we have

$$|\,|\psi\rangle\,|^2 = (|\psi\rangle)^\dagger\,|\psi\rangle$$

by analogy with the real case. In general, we then have that

$$\langle\psi_1|\psi_2\rangle = \langle\psi_1, \psi_2\rangle$$

so the combination of a bra and a ket is a bracket. Ha ha ha. In particular, we have

$$\langle\psi|\psi\rangle = |a|^2 + |b|^2$$

Since we want the total probability of measuring something to be 1, we want to enforce

$$\langle\psi|\psi\rangle = 1$$

It is therefore natural to identify $|a|^2$ with the probability of measuring $|L\rangle$ and $|b|^2$ the probability of measuring $|R\rangle$. From this, we can derive the *Born rule*.

$$P(\text{measuring state}\,|A\rangle) = |\,\langle A|\psi\rangle\,|^2$$

If you didn't believe me, we can compute explicitly

$$P(L) = |\,\langle L|\psi\rangle\,|^2 = |a|^2 \qquad P(R) = |\,\langle R|\psi\rangle\,|^2 = |b|^2$$

The strange thing about quantum states is that they "collapse" after measurement. That is, if $|L\rangle$ is measured, the state immediately goes from being $|\psi\rangle$ to $|L\rangle$. Mathematically, this is represented by a projection operator

$$P\,|\psi\rangle \propto (|L\rangle\,\langle L|)\,|\psi\rangle$$

This *decoherence* is a huge headache for quantum computation, and it has some very counterintuitive consequences, as we'll see later.

It might seem tempting to somehow canonically identify

$$|\psi\rangle \sim ? \begin{bmatrix} a \\ b \end{bmatrix}$$

I did say it was a vector space, after all, and some hastily-made linear algebra primers define a vector space as "a list of numbers". However, this is a mistake because everything is dependent on a choice of *basis*. We're currently working in the "position basis". If we switch to what I'm going to call (and what no one else calls) the "momentum basis"

$$|p_0\rangle = \frac{1}{\sqrt{2}}\,|L\rangle + \frac{1}{\sqrt{2}}\,|R\rangle$$
$$|p_1\rangle = \frac{1}{\sqrt{2}}\,|L\rangle - \frac{1}{\sqrt{2}}\,|R\rangle$$

then suddenly our state looks very different

$$|\psi\rangle = \frac{a+b}{\sqrt{2}}|p_0\rangle + \frac{a-b}{\sqrt{2}}|p_1\rangle$$

(Verify for yourself that this is the same state!) Performing a "momentum measurement" would then collapse the state into either $|p_0\rangle$ or $|p_1\rangle$ and not our old basis states.

Observables. What determines a basis? Remember that when do a measurement, we're measuring *something*. Let's recall how measurements work in the classical case. A classical observable is just a function

$$O : \{\text{classical states}\} \to \mathbb{R}$$

We're allowed to do this because classical states always have a well-defined position, momentum, energy, etc. However, this isn't the case in quantum mechanics. It is the case, however, that for every observable we can write our quantum state as a superposition of states with well-defined values. Definite states of an observable therefore always form an orthonormal basis. Since we want our observable to respect linearity, we can define instead of the classical definition a linear map

$$\hat{O} : \{\text{quantum states}\} \to \{\text{quantum states}\}$$

whose action on the basis just scales the state vector by the value of the observable. For example, with the position operator

$$\hat{x}|x=2\rangle = 2|x=2\rangle$$

or the momentum operator

$$\hat{p}|p=-5\rangle = -5|p=-5\rangle$$

In this way, we can recover the classical function O by taking $\langle\psi|\hat{O}|\psi\rangle$ on definite states

$$\langle x=2|\hat{x}|x=2\rangle = 2$$

Observables in quantum mechanics are therefore linear maps (matrices) with a orthonormal basis of eigenvectors with real eigenvalues. This implies $\hat{O}$ is *Hermitian*.

Definition 1.2. A matrix $\hat{O}$ is Hermitian if $\hat{O}^\dagger = \hat{O}$

Key Idea 1.3. In quantum mechanics, observables are Hermitian operators on the Hilbert space of quantum states.

As it turns out (and this is nontrivial), in finite dimensions, which is what we like to work with in quantum computing, this is an "if and only if"

Theorem 1.4. *(the Spectral Theorem) Over a finite-dimensional Hilbert space, the following are equivalent*

- $\hat{O}^\dagger = \hat{O}$
- $\hat{O}$ *has an orthonormal basis of eigenvectors with real eigenvalues*

Remark 1.5. (For mathematicians only, physicists please skip). No one seems to care that it's only an equivalence when the Hilbert space is finite dimensional. For example, the real-deal position operator

$$\hat{x} : L^2(\mathbb{R}) \to L^2(\mathbb{R}) \qquad \psi(x) \mapsto x\psi(x)$$

doesn't have any eigenvectors in $L^2(\mathbb{R})$ at all! Physicists like to pretend it has a "position basis" of delta functions $\delta(x-\alpha)$, but these aren't in $L^2(\mathbb{R})$! Fortunately, it's not much of an actual problem when it comes to doing quantum mechanics.

In the abstract theory of quantum computation, 90% of the time we work in the *computational basis*, which is typically a curated subspace of the energy operator H we can use to store information (more on this next week). However, it is still useful to be able to think in different bases, e.g. the *Bell basis*

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle) \qquad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$$

But enough yapping.

The Stern-Gerlach Experiment. Let's see how all of this actually works.

In 1922, it was discovered that silver atoms have a property called *spin* which causes the atoms to split between "up" and "down" in a certain magnetic field (henceforth "Stern-Gerlach apparatus").[1] If we measure the spin along the z-axis we can therefore write the quantum state of the silver atoms

$$|\psi\rangle = a\,|z_\uparrow\rangle + b\,|z_\downarrow\rangle$$

If we put $|\psi\rangle$ through a z-axis Stern-Gerlach apparatus, we get that $|a|^2$ of the silver atoms will be in the $|z_\uparrow\rangle$. Let's take all of these atoms and put them through an x-axis Stern-Gerlach apparatus. As it turns out, $|z_\uparrow\rangle$ atoms will split into $|x_\uparrow\rangle$ and $|x_\downarrow\rangle$ with equal probability. We can explain this using a new basis

$$|z_\uparrow\rangle = \frac{1}{\sqrt{2}}\,|x_\uparrow\rangle + \frac{1}{\sqrt{2}}\,|x_\downarrow\rangle \qquad |z_\downarrow\rangle = \frac{1}{\sqrt{2}}\,|x_\uparrow\rangle - \frac{1}{\sqrt{2}}\,|x_\downarrow\rangle$$

The minus sign is there because the z states need to form a basis. Similarly, if we have a y-axis Stern-Gerlach apparatus, then both $|z_\uparrow\rangle$ and $|x_\uparrow\rangle$ will split into $|y_\uparrow\rangle$ and $|y_\downarrow\rangle$ with equal probability. It's not possible to write down such a basis using only real numbers, and this is ultimately the reason we have to use complex-valued coefficients

$$|y_\uparrow\rangle = \frac{1}{\sqrt{2}}\,|z_\uparrow\rangle + \frac{i}{\sqrt{2}}\,|z_\downarrow\rangle \qquad |y_\downarrow\rangle = \frac{1}{\sqrt{2}}\,|z_\uparrow\rangle - \frac{i}{\sqrt{2}}\,|z_\downarrow\rangle$$

What's weird is that if we send a stream of $|z_\uparrow\rangle$ through an x-axis Stern-Gerlach apparatus, keep the ones that go to $|x_\uparrow\rangle$, and then put them through a z-axis apparatus, we get an even mix of $|z_\uparrow\rangle$ and $|z_\downarrow\rangle$ atoms, even though there were no $|z_\downarrow\rangle$ atoms to begin with! Even freakier, if we instead send both streams of atoms out of the x apparatus into the z apparatus (in particular, if we don't perform a measurement), then 100% of the atoms will be $|z_\uparrow\rangle$, even though we've only added more atoms!

Quantum mechanics is weird. Verify all of the above using the mathematical techniques you've learned and work through the computations until you really understand it. Remember,

> "Mathematics is not a spectator sport!"
>
> (Prof. Lek-Heng Lim, STAT 28000)

[1] This isn't actually true. Spin wasn't discovered until 1924 and Stern and Gerlach thought they had found something else.

2. (Alice) Qubits and Architectures

Qubits. Now that you know about quantum states, we build up the definition for a *qubit*. This is the fundamental logical unit used in quantum computing. Recall that in classical computing, the corresponding fundamental units are *bits*. These take on the value of either 1s or 0s (i.e. encoded in binary), in other words, "true" or "false" respectively.

From the previous lecture, we have an orthonormal basis made up of quantum states. In quantum computing, these basis vectors represent our new notion of 1 and 0 or true and false, and we can write superpositions between them. More specifically, what I mean is:

- Assign some basis vector (say, "spin-up" $|+z\rangle$) to the vector $|0\rangle$.
- Similarly, assign another basis vector (say, "spin-down" $|-z\rangle$ to the vector $|1\rangle$.
- Then, any qubit can be written as some state $|\psi\rangle = a\,|z+\rangle + b\,|z-\rangle = a\,|0\rangle + b\,|1\rangle$ (with rules for a and b as covered in Lecture 1).

For computational purposes we identify qubits up to *global phase*, since it's impossible to distinguish between states that are otherwise the same.

$$|\psi\rangle \sim e^{i\theta}\,|\psi\rangle$$

But why do we even want to define a qubit in the first place? The answer boils down to two main properties that classical bits *don't* have (but which are contained in quantum mechanics!):

(1) **Superposition**: This property increases efficiency since it allows a quantum computer to explore many possibilities at once (*quantum parallelism*).
(2) **Entanglement**: We'll cover this more in Lecture 4, but this basically means that we can exploit the phenomenon of quantum entanglement to increase computational power.

This means that using qubits, we can solve problems that are very hard or impossible with classical bits, such as:

- Searching through a database (Grover's)
- Factoring prime numbers (Shor's)
- Certain optimization problems (i.e. many things)
- Simulation of quantum systems (duh).

We will elaborate more on this in Lecture 5.

The Bloch sphere. A natural representation of qubits is *the Bloch sphere*. It is shown in Figure 1.

Observe that the positive and negative z-axis represent the orthonormal basis vectors $|0\rangle$ and $|1\rangle$, respectively. The positive and negative x and y axes represent other orthonormal basis vectors corresponding to 90° rotations of $|0\rangle$ and $|1\rangle$ (you have already seen a form of this in the previous lecture's discussion of the Stern-Gerlach experiment). Using such a coordinate system, any qubit on the surface of the Bloch sphere is called a *pure state*, and can be uniquely written as

$$|\psi\rangle = \cos\left(\theta/2\right)|0\rangle + e^{i\varphi}\sin\left(\theta/2\right)|1\rangle\,.$$

It should not be too difficult to check for yourself geometrically why this is true (hint: it comes pretty directly from spherical coordinates). Observe that the polar

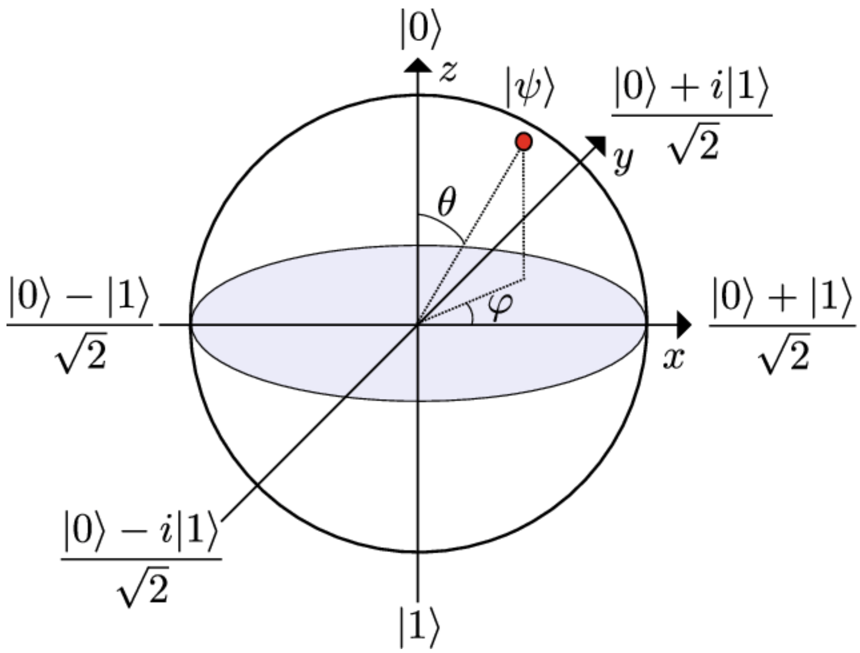


FIGURE 1. The Bloch sphere (stolen from the internet, but it's okay because they stole from Bloch).

angle θ controls the probabilities of $|0\rangle$ and $|1\rangle$, whereas the azimuthal angle φ is something called the *relative phase*. Also notice that points within the Bloch sphere represent some probabilistic mixture of pure states— that is, they are *mixed states*.

Qubit architectures. Again, recall classical bits. These encoded 0s and 1s don't just exist in a vacuum— they have to be implemented through the physical architecture of the computer. In the classical case, this is done through transistors in a CPU. In the quantum case, there are a few main options (as follows).

Superconducting qubits. These qubits are made out of superconducting materials (i.e. materials with zero resistance at very low temperatures). They exploit the *Josephson junction*, in which electrons quantum tunnel through an insulator between layers of superconductors, to create a sort of "artificial atom" with discrete energy levels. Of these, there are two types:

- The *transmon* (the original superconducting qubit) is a Josephson junction connected in parallel to a capacitor, which provides a high degree of robustness against noise (i.e. longer *coherence times*). This is at the expense of the *anharmonicity*.
- The *fluxonium qubit* connects a Josephson junction to a superinductor that modifies the potential of the transmon to increase anharmonicity and resistance to noise.

Photonic qubits. These attempt to encode the information of qubits through the properties of photons. For example:

- Encoding $|0\rangle$ and $|1\rangle$ as different (orthogonal) polarization directions.

- Creating two possible waveguides corresponding to $|0\rangle$ and $|1\rangle$ for the photon to travel down (*dual-rail*).

Photonics-based quantum systems are naturally noise-resistant and are in some sense more accessible since they operate at room temperature using integrated photonics. However, they are very annoying (speaking from experience) and more broadly, interactions between photons are difficult to engineer.

Trapped ions. These are literally just ions suspended by electromagnetic fields and laser-cooled for stability. Then the qubit's $|0\rangle$ and $|1\rangle$ correspond directly to the ion's discrete energy levels (tuned by lasers). As you might infer, this is very resistant to noise (*high-fidelity*), but gate operations can be slow and these systems are difficult to work with.

Neutral atoms. These are very similar to the trapped ion qubits, but use neutral atoms instead of ions (obviously)- in particular, they use Rydberg atoms excited to very high energy levels. The main difference from the trapped ion case is that they are trapped using optical tweezers rather than EM fields. The trade off here is that while neutral atoms are easier to manipulate, they are much more susceptible to noise.

Topological qubits. These qubits are the most mathematically/theoretically interesting but also the least experimentally-realized. The idea is that in some condensed matter systems there arise these quasiparticles called *anyons*, which have inherent topological properties that can be used to encode a $|0\rangle$ and $|1\rangle$ state. We will likely talk about these in more detail in Lecture 6.

Tensor product of qubits. So far, you have learned about the mathematical representation of a single qubit. What happens when we combine these into multiple-qubit systems? Mathematically, we use the notion of a *tensor product*. In quantum mechanics, for any two states $|\psi_1\rangle$ and $|\psi_2\rangle$, we can write this as

$$|\psi_1\rangle \otimes |\psi_2\rangle = |\psi_1\psi_2\rangle \,.$$

This allows us to treat the interaction between the two states as its own quantum state! In fact, we can take the tensor product of any n number of states to build up a single quantum state. Note that this is a *bilinear map*, i.e.

$$(a\,|\psi_1\rangle + b\,|\psi_2\rangle) \otimes |\psi_3\rangle = a\,|\psi_1\psi_3\rangle + b\,|\psi_2\psi_3\rangle$$

(and likewise on the other side). As an example in quantum computing specifically, consider a 2-qubit system. Recall that each qubit has basis states $\{|0\rangle\,, |1\rangle\}$. Based on this, we can assign the following states to each qubit:

- first qubit: $|0\rangle_1$ and $|1\rangle_1$,
- second qubit: $|0\rangle_2$ and $|1\rangle_2$.

Together, the two qubits form a new vector space with the orthonormal basis

$$\{|00\rangle := |0\rangle_1 \otimes |0\rangle_2\,, |10\rangle := |1\rangle_1 \otimes |0\rangle_2\,, |01\rangle := |0\rangle_1 \otimes |1\rangle_2\,, |11\rangle := |1\rangle_1 \otimes |1\rangle_2\}.$$

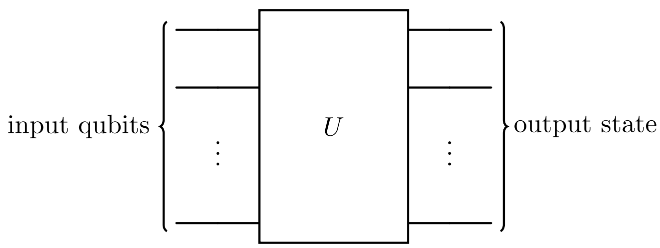
Furthermore, operators on this fancy-schmancy 4-dimensional vector space can be written as the tensor product of operators on the single-qubit states. For example, the identity is just $I := I_1 \otimes I_2$.

3. (Ryan) Quantum Circuits

3.1. Unitarity.

Definition 3.1. A *unitary operator* (matrix) satisfies $U^\dagger = U^{-1}$ (equivalently $U^\dagger U = I$)

Fundamentally, every "quantum algorithm" is just this:



We start with some set of qubits, apply a unitary operator to them, and then get some output state of the qubits. The fundamental reason for this is that *everything* in life behaves like this, not just qubits. To see why, we can look at the equation governing time evolution in quantum mechanics, the *Schrödinger equation.*

$$\frac{\partial}{\partial t} |\psi(t)\rangle = -\frac{i}{\hbar}\hat{H} |\psi(t)\rangle$$

where $\hat{H}$ is the *Hamiltonian* (a fancy word for "energy"). The easiest way to solve a differential equation is to write down the answer. If $\hat{H}$ does not depend on time, we can write a formal solution by defining a "time evolution operator"

$$T(t) := e^{-\frac{i}{\hbar}\hat{H}t}$$

where we define the *matrix exponential*

$$e^A := \sum_{n=0}^{\infty} \frac{A^n}{n!}$$

Then basically by definition

$$|\psi(t)\rangle = T(t) |\psi(0)\rangle$$

solves the Schrödinger equation. (Verify this yourself!) Likewise, if $\hat{H}$ does depend on time, we can define

$$T(t) := e^{-\frac{i}{\hbar}\int_0^t H(t')dt'}$$

and get the same result. We'll work with the time-independent case because it's slightly easier, but everything extends naturally.

If we could easily compute T, physicists wouldn't have a job. Fortunately, working with this definition is a pain. However, we can still determine some properties of a system. For example, look what happens if we compute

$$T^\dagger = \left(e^{-\frac{i}{\hbar}\hat{H}t}\right)^\dagger = e^{+\frac{i}{\hbar}\hat{H}^\dagger t}$$

Since $\hat{H}$ is an observable, it is a Hermitian operator, so $\hat{H}^\dagger = \hat{H}$. Moreover, since

$$e^{-\frac{i}{\hbar}\hat{H}t}e^{\frac{i}{\hbar}\hat{H}t} = I$$

(exactly like ordinary exponentiation) we have that

$$T^\dagger T = I$$

so T is unitary! Therefore, any possible evolution of quantum states must be unitary, including any quantum algorithm.

3.2. **Quantum gates.** But how do we implement actual unitary operators on real qubits? In general, it's much easier to break an arbitrary unitary down into a small set of *quantum gates*, each of which can be implemented on hardware.[2]

One common gate is the *Hadamard gate*

$$\begin{aligned} |0\rangle &\mapsto \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \\ |1\rangle &\mapsto \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \end{aligned}$$

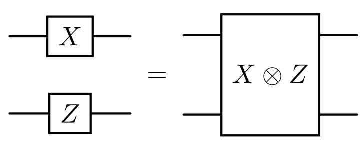
We can represent this as a unitary matrix

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Also common are the *Pauli gates*.

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

which show up all over quantum mechanics. We can construct two-qubit gates using the funny-looking *Kronecker product*.



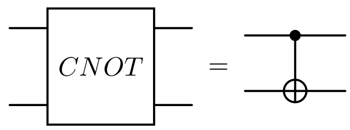
$$X \otimes Z = \begin{bmatrix} 0\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} & 1\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ 1\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} & 0\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$$

Verify for yourself that, for example,

$$(X \otimes Z)|01\rangle = -|11\rangle = (X|0\rangle) \otimes (Z|1\rangle)$$

There are other two-qubit gates, though. One common enough to have its own symbol is the CNOT ("controlled-NOT") gate.

[2]An alternative and more interesting breakdown is given by the *ZX-calculus*, of particular interest to computer scientists, mathematicians, and anyone else who never has to touch physical hardware in their lives.



$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

One of the reasons for the focus on CNOT is that single-qubit gates and CNOT form a *universal gate set*, which means that any unitary on any number of qubits can be approximated to any precision with just these gates. The study of universal gate sets is interesting in its own right but too technical to get into here.

3.3. **The "No-Cloning" Theorem.** Now that we've seen what quantum computers *can* do, let's see what quantum computers *can't* do.

One incredibly important feature of unitary matrices is that they preserve the inner product. We can simply compute

$$\langle U\psi_1 | U\psi_2\rangle = (U|\psi_1\rangle)^\dagger (U|\psi_2\rangle) = \langle\psi_1| U^\dagger U |\psi_2\rangle = \langle\psi_1|\psi_2\rangle$$

This is great for time evolution, since it means that the time evolution of a normalized state is still a normalized state. However, it lets us prove

Theorem 3.2. *(the No-Cloning Theorem) There does not exist a unitary U such that*

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$$

for all states $|\psi\rangle$.

In plain English, it's impossible to copy quantum states.

Proof. Assume there existed such a U. Pick arbitrary quantum states $|\psi_1\rangle$, $|\psi_2\rangle$. Then

$$\begin{aligned} \langle\psi_1|\psi_2\rangle &= (\langle\psi_1| \otimes \langle 0|)(|\psi_2\rangle \otimes |0\rangle) = (\langle\psi_1| \otimes \langle 0|)U^\dagger U(|\psi_2\rangle \otimes |0\rangle) \\ &= (\langle\psi_1| \otimes \langle\psi_1|)(|\psi_2\rangle \otimes |\psi_2\rangle) = (\langle\psi_1|\psi_2\rangle)^2 \end{aligned}$$

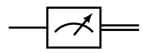
so $\langle\psi_1|\psi_2\rangle$ must be either 0 or 1. This tells us that $|\psi_1\rangle$ and $|\psi_2\rangle$ must be basis states (classical bits) and not arbitrary qubits. $\square$

There are strengthenings of this statement to include arbitrary phase factors

$$U(|\psi\rangle \otimes |0\rangle) \neq e^{i\alpha} |\psi\rangle \otimes |\psi\rangle$$

but fundamentally the problem is that unitary matrices are always *reversible* (i.e. U^{-1} always exists), which means that it's impossible to create or destroy information with them.

3.4. **The Measurement Problem.** We have one last gate to explore, the *measurement gate.*



Unlike the other gates, this takes in a qubit and returns a classical bit depending on the outcome of the measurement.

We have a problem, though: measurement is *not* reversible. How does this work with unitary time evolution? This depends on your *interpretation* of quantum mechanics. Although quantum mechanics as a *theory* is not debated, there is debate as to what the different aspects of the theory actually represent.

The most popular interpretation by far is the *Copenhagen interpretation*, which says that evolution by the Schrödinger equation and measurement are two separate processes. In essence, add "except measurement" to everything I've said so far and it remains valid.

An alternate interpretation which I only mention because of its prevalence in pop-science media is the *many-worlds interpretation.* This is the kernel of truth that Google used to (entirely falsely) claim that "quantum computing is evidence of parallel universes". What's really going on is the simple observation that non-unitary operators can be seen as the restriction of unitary operators on a larger Hilbert space. For example, a "measurement" in this framework is really just the unitary map

$$|\psi\rangle\,|\text{scientist}\rangle \mapsto a\,|0\rangle\,|\text{scientist measures } 0\rangle + b\,|1\rangle\,|\text{scientist measures } 1\rangle$$

This general trick for dealing with non-unitary evolution (e.g. as it occurs in open quantum systems) got the nickname "the Church of the Larger Hilbert Space". The many-worlds interpretation holds that "measurement" as in the Copenhagen interpretation doesn't exist and that the universe is always in this complicated superposition of all possible measurement outcomes that could have ever taken place. A further extension (not core to the interpretation) posits that the different states in this superposition are different "parallel universes", and this and *only* this is the reason people claim any connection between quantum mechanics and "the multiverse". Additionally, this interpretation struggles to explain the Born rule or even more generally why we never "feel" this superposition, so there's no reason it should be this prominent other than that it generates clicks for headlines.

4. (Rick) Quantum Entanglement

While it remains a mystery, our ability to describe the phenomena of quantum entanglement allows us to take advantage of quantum entanglement as a powerful resource for various tasks, including but not limited to *quantum communication* to *measurement-based quantum state preparation.* Here, we'll go over the notion of quantum entanglement, go through a few examples, and cover quantum teleportation.

4.1. "Physical" Interpretation of Entanglement.

Before going through how we describe quantum entanglement mathematically, it is important to understand what quantum entanglement mean physically; more specifically, how it differs from classical correlation.

Consider two quantum systems (i.e., both are in a quantum superposition of various states), A and B. Quantum entanglement refers to the following phenomenon: suppose A is entangled with B. Without loss of generality, if we choose to perform a quantum measurement on system A (causing the superposition of A to collapse to a single state), the quantum state of system B consequently collapses to a corresponding state determined by the entanglement between A and B. Although this correlation appears instantaneous, no information is transmitted faster than light; rather, **the two systems share a single joint quantum state**. This is very strange as one cannot predict what state either system will collapse to before measurement, and Einstein referred to entanglement as a "spooky action at a distance".

In both classical and quantum systems, two subsystems (say, Alice's and Bob's) can have *correlations*; meaning that if you measure one, it gives you information about the other. But **how** that correlation arises is completely different from one another, which is related to how the probability distribution for each subsystem is described.

- *Classical correlation* arises from shared randomness or a common cause: for instance, consider a box containing one red ball and one blue ball. If Alice takes one ball first and Bob takes the remaining second ball, the probability distribution that describes which ball each person took is a classical probability distribution and classically correlated. However, there is no quantum coherence between the two alternatives.
- Converse, *quantum entanglement (correlation)* arises from quantum superposition across composite systems. For the entangled state, the correlations persist in every basis. That's the quantum signature: the correlations are basis-independent and cannot be explained by shared randomness.

Now, let's move onto the mathematical description of quantum entanglement.

4.2. **On Tensor Products of Hilbert Spaces.** Before going into the mathematical description of quantum entanglement, it will be useful to go into the axioms of tensor products, which will serve as a foundation for all mathematical operations we will do during this and future lectures that involve tensor products (which is guaranteed to happen, since describe many-body quantum systems requires tensor products).

In quantum mechanics, systems composed of two subsystems (or two degrees of freedom) are referred to as **bipartite states**. By the postulates of quantum mechanics, such composite systems are described as vectors in the **tensor product space**

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2,$$

where $\mathcal{H}_1$ and $\mathcal{H}_2$ are the Hilbert spaces associated with the two subsystems. In this discussion (and pretty much all other discussion unless otherwise specificed), we will focus on the finite-dimensional case.

Recall that a **tensor product** can be intuitively be thought of as a means to "stitch" two (or multiple) vector spaces or Hilbert spaces over a field $\mathbb{F}$ together to form a new Hilbert space. Thus, given Hilbert spaces $\mathcal{H}_1$, $\mathcal{H}_2$, the tensor product $\otimes : \mathcal{H}_1 \times \mathcal{H}_2 \to \mathcal{H}_1 \otimes \mathcal{H}_2$ gives a new Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$. Moreover, since $\otimes$ is a bilinear map, we have the following properties:
for all $|\psi_1\rangle \in \mathcal{H}_1$, $|\psi_2\rangle, |\psi_2'\rangle \in \mathcal{H}_2$, $k \in \mathbb{F}$

- $k(|\psi_1\rangle) \otimes |\psi_2\rangle = k(|\psi_1\rangle \otimes |\psi_2\rangle) = |\psi_1\rangle \otimes (k\,|\psi_2\rangle)$
- $|\psi_1\rangle \otimes ((|\psi_2\rangle) + (|\psi_2'\rangle)) = (|\psi_1\rangle) \otimes |\psi_2\rangle) + (|\psi_1\rangle) \otimes |\psi_2'\rangle)$
- $((|\psi_2\rangle) + (|\psi_2'\rangle)) \otimes |\psi_1\rangle = (|\psi_1\rangle) \otimes |\psi_2\rangle) + (|\psi_1\rangle) \otimes |\psi_2'\rangle)$

The last property will be particularly useful, since it will allow to do computations as such:

$$(|0\rangle + |1\rangle)(|0\rangle + |1\rangle) = |00\rangle + |01\rangle + |10\rangle + |11\rangle$$

where we've used the tensor product shorthand for ket notation:

$$|i\rangle \otimes |j\rangle = |i\rangle\,|j\rangle = |ij\rangle$$

Now, we can mathematically define what an entangled quantum state is!

4.3. **Bipartite Systems.** For now, we will focus on two-body entanglement, i.e. entanglement of a bipartite system. The study of bipartite systems is central to quantum information theory because **two-body entanglement** is both the most fundamental and best understood type of quantum correlation. Many key quantum phenomena, such as Bell inequalities, teleportation, and superdense coding, can be fully explained in the bipartite framework. Conversely, higher-order multipartite entanglement (involving three or more subsystems), sometimes referred to as **many-body entanglement**, is more complex and less well classified, so bipartite entanglement serves as the conceptual foundation. That being said, we'll touch on multipartite entanglement in the future, as they are fascinating in their own right.

4.4. **Entangled vs Separable (Product) States.** Now we can formally start discussing quantum entanglement!

Here is our setup: let $\{|a_i\rangle\}_{i=1}^n \subset \mathcal{H}_1$ and $\{|b_j\rangle\}_{j=1}^m \subset \mathcal{H}_2$ be orthonormal bases for $\mathcal{H}_1$ and $\mathcal{H}_2$, respectively. Physically, these basis states represent the possible physical (eigen)states of some observable that can be measured; in the context of quantum computing and information, they are often taken to be the computational basis states.

Then, the set $\{|a_i\rangle \otimes |b_j\rangle\} = \{|a_i b_j\rangle\}$ forms a basis for the composite Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$.

Any normalized vector (i.e., any pure state) in the composite system can be written as

$$|\psi\rangle = \sum_{i,j} c_{ij} \big(|a_i\rangle \otimes |b_j\rangle\big) = \sum_{i,j} c_{ij} |a_i b_j\rangle,$$

where c_{ij} are complex coefficients satisfying $\sum_{i,j} |c_{ij}|^2 = 1$, which correspond to the probability amplitudes. Now, we can define *entangle*d states!

Definition 4.1. (Entangled states vs Product states)
Let $|\psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$ be a quantum state in a bipartite system (for instance, a system of two qubits). If the state $|\psi\rangle$ can be expressed as

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$$

with $|\psi_1\rangle \in \mathcal{H}_1$ and $|\psi_2\rangle \in \mathcal{H}_2$ being pure states of the individual subsystems, then $|\psi\rangle$ is called a **product state**, or equivalently, a **separable state** (technically a state is separable if it can be expressed as a convex combination of product states, but that is not important here).

Otherwise, if no such factorization exists, the state is called **entangled**.

Physical intuition: A product (separable) state describes two subsystems that are completely independent — measuring one gives no information about the other. An entangled state, on the other hand, exhibits correlations that cannot be explained by any local description: the subsystems do not possess independent physical reality, and their properties are intrinsically linked by the structure of the joint quantum state.

Note that for pure states, the notions of "product" and "separable" coincide. However, this equivalence breaks down for **mixed states**, where one must use density matrices and the concept of *separable mixtures* to distinguish classical from quantum correlations.

Now, let us discuss some examples of entangled states and product states (states with no entanglement).

4.5. **The Paragon of Entangled States: Bell Pairs.** Among all bipartite entangled states, none are more fundamental than the **Bell states** (also known as the *EPR pairs*). These are the maximally entangled two-qubit pure states, forming an orthonormal basis for $\mathbb{C}^2 \otimes \mathbb{C}^2$. In other words, the Bell pairs (which there are four of them, as you'll see) span the space of all possible two-qubit states!

In most of the literature, the Bell states are written as

$$|\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \qquad |\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle).$$

These four states are mutually orthogonal, and every two-qubit state can be expanded in this basis.

However, you could ask, what do we mean by *maximally entangled?* A **maximally entangled state**, such as a Bell pair, is a state that contains the *maximum possible amount of correlation* allowed between two qubits; this is one we physically mean by "maximally entangled". For example, in the Bell state

$$|\mathbb{1}\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}),$$

neither A nor B has a definite state of its own. Each subsystem is completely mixed:

$$\rho_A = \rho_B = \frac{\mathbb{1}}{2}.$$

All the information about the joint system is stored entirely in the *correlations* between A and B. Since each subsystem is already as uncertain as possible, there is "no room left" for either qubit to share additional quantum correlations with a third system.

A convenient operator-label notation. There is a particularly useful way to organize the Bell states that emphasizes their symmetry under Pauli operators. Consider the canonical Bell state

$$|\mathbb{1}\rangle \; := \; \frac{1}{\sqrt{2}}\big(\,|00\rangle + |11\rangle\,\big),$$

which one may view as the "identity Bell state" because all other Bell states can be obtained from it by acting with single-qubit Pauli operators.

We will adopt the convention that single-qubit Pauli operators act on the *first* qubit of the Bell pair unless otherwise specified. We will also write states such as $|00\rangle_{23}$ or $|\psi\rangle_1$ to indicate explicitly which qubits they refer to, the former referring to qubits two and three, and the latter referring to qubit 1 (a single qubit state).

Starting from the canonical Bell state

$$|\mathbb{1}\rangle_{12} := \frac{1}{\sqrt{2}}\big(\,|00\rangle_{12} + |11\rangle_{12}\,\big),$$

the remaining Bell states are obtained by applying X, Z, and ZX to the first subsystem:

$$|X\rangle_{12} = (X \otimes \mathbb{1})\,|\mathbb{1}\rangle_{12}\,, \qquad |Z\rangle_{12} = (Z \otimes \mathbb{1})\,|\mathbb{1}\rangle_{12}\,, \qquad |ZX\rangle_{12} = (ZX \otimes \mathbb{1})\,|\mathbb{1}\rangle_{12}\,.$$

Explicitly,

$$|\mathbb{1}\rangle_{12} = \frac{1}{\sqrt{2}}\big(\,|00\rangle_{12} + |11\rangle_{12}\,\big),$$

$$|X\rangle_{12} = \frac{1}{\sqrt{2}}\big(\,|10\rangle_{12} + |01\rangle_{12}\,\big),$$

$$|Z\rangle_{12} = \frac{1}{\sqrt{2}}\big(\,|00\rangle_{12} - |11\rangle_{12}\,\big),$$

$$|ZX\rangle_{12} = \frac{1}{\sqrt{2}}\big(\,|10\rangle_{12} - |01\rangle_{12}\,\big).$$

Make sure to confirm this for yourself!

Alternative notation using subscripts. When it is helpful to emphasize which qubit an operator acts on, we will use a subscript indicating its target:

$$X_1 = X \otimes \mathbb{1}, \qquad Z_1 = Z \otimes \mathbb{1}, \qquad X_2 = \mathbb{1} \otimes X, \qquad Z_2 = \mathbb{1} \otimes Z.$$

With this notation the Bell states become

$$|X\rangle_{12} = X_1\,|\mathbb{1}\rangle_{12}\,, \qquad |Z\rangle_{12} = Z_1\,|\mathbb{1}\rangle_{12}\,, \qquad |ZX\rangle_{12} = (Z_1X_1)\,|\mathbb{1}\rangle_{12}\,.$$

Finally, this labeling is consistent (up to irrelevant global phases) with the standard Bell-state notation:

$$|\mathbb{1}\rangle = |\Phi^+\rangle\,, \qquad |Z\rangle = |\Phi^-\rangle\,, \qquad |X\rangle = |\Psi^+\rangle\,, \qquad |ZX\rangle = |\Psi^-\rangle\,.$$

I want to emphasize that the set of Bell pairs

$$\{|\mathbb{1}\rangle\,, |X\rangle\,, |Z\rangle\,, |ZX\rangle\}$$

form an orthonormal basis for the (Hilbert) space of all two-qubit states; because this basis is so commonly used and important, we refer to this basis as the **Bell basis.**

Why this labeling is useful. The Pauli-labeled basis makes Bell states especially convenient in quantum information protocols. Any Bell measurement outcome corresponds precisely to detecting which Pauli error has been applied to one qubit relative to the reference Bell pair. This will become essential in the description of **quantum teleportation**, where Bob corrects his qubit by applying the appropriate Pauli operator indexed by the Bell measurement outcome.

4.6. **The Monogamy of Entanglement.** Entanglement behaves very differently from classical correlation. One of the most striking differences is a structural constraint known as the **monogamy of entanglement**. Informally:

> *If system A is maximally entangled with system B, then A cannot be entangled at all with any third system C.*

This is in sharp contrast to classical correlations, which can be freely shared among many parties. Let's consider an example involving Bell pairs.

A simple three–qubit example. Suppose Alice and Bob share the Bell pair $|\mathbb{1}\rangle_{AB}$, where the index AB corresponds to one qubit owned by Alice and Bob.

Now imagine a third qubit C held by Charlie. Could qubit A also be entangled with C?

If that were possible, then A would simultaneously be entangled with both B and C. But this would mean A has two different "partners" sharing information about its quantum state. However, in a Bell state, qubit A has *no state of its own*:

$$\rho_A = \frac{\mathbb{1}}{2} \quad \text{(maximum entropy)}.$$

If A were entangled with C, some of the uncertainty in A must now come from correlations with C, not just with B. This would imply that the AB pair is no longer a pure Bell pair.

But we started with AB *maximally* entangled. Thus the presence of C can only be consistent if

$$C \text{ is completely uncorrelated with } A \text{ and } B.$$

In other words, the only state compatible with AB being a true Bell pair is a product state of the form

$$|\mathbb{1}\rangle_{AB} \otimes |\phi\rangle_C .$$

Pure-state intuition. This explanation involves the use of density matrices, so feel free to skip over this section and come back later once you are more familiar with density matrices.

Suppose AB is in a pure state $|\psi_{AB}\rangle$. If A is maximally entangled with B, then the reduced density matrix of A is maximally mixed:

$$\rho_A = \mathrm{Tr}_B(|\psi_{AB}\rangle \langle\psi_{AB}|) = \frac{\mathbb{1}}{2}.$$

Now imagine introducing a third system C. If A had any correlation with C, the joint state AB would not remain pure; part of the "uncertainty" in A would be shared with C instead of B. But maximal entanglement already saturates the entropy allowed for A, leaving no "room" for entanglement with any other system.

Formal statement. For three qubits A, B, C, the Coffman–Kundu–Wootters monogamy relation states that

$$\tau_{A:BC} \;\geq\; \tau_{A:B} + \tau_{A:C},$$

where τ is the *tangle*, a measure of bipartite entanglement. When A and B share a maximally entangled Bell pair ($\tau_{A:B} = 1$), the inequality forces

$$\tau_{A:C} = 0.$$

Operational meaning. This impossibility of "sharing" maximal entanglement underlies the security of quantum key distribution, prevents cloning of entanglement, and governs the structure of multipartite quantum states such as GHZ and W states. In short: entanglement is an *exclusive* resource.

4.7. **Quantum Teleportation.** Quantum teleportation is a landmark protocol demonstrating how entanglement and classical communication allow one to transfer an *unknown quantum state* from one location to another without physically sending the particle itself.

In all contexts of communication, i.e. sending information from one location to another, a necessary component of communication is a physical medium through which the information is encoded in and travels through. For instance, during wars where electronic signals were not accessible technology, fire was used to signal some piece of information, e.g. that the opponents were on their way. To be discrete, the information encoded in the photons (from the fire) would travel through space and enter the solider's retinas.

What's amazing about quantum teleportation is that there is no physical medium for which the quantum bit of information travels through, becausse it doesn't: this is why we refer to this protocol as *teleportation*! The catch however, is that two pieces of classical information (bits) are transferred instead, which captures the "essence" of the quantum bit of information we're teleporting.

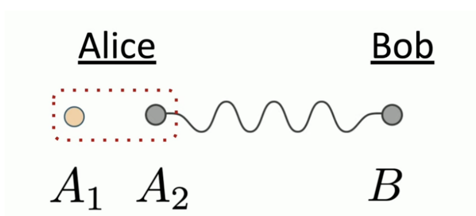
Setup. Alice holds an unknown qubit

$$|\psi\rangle_{A_1} = \alpha\,|0\rangle + \beta\,|1\rangle\,,$$

which is a quantum bit of information that Alice would like to send, or "teleport" to Bob. Morover, she and Bob share a Bell pair, say the canonical one,

$$|\mathbb{1}\rangle_{A_2 B} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Alice possesses qubits A_1 (the unknown state) and A_2 (half of the Bell pair), while Bob possesses qubit B. Note that A_1, A_2, B are just labels for these three qubits that are involved in this protocol. Here is a visualization of our initial set-up:



Before going behind the full mathematics that allows this to work, here is a **qualitative description** of the quantum teleportation protocol.
To reiterate, setup is that Alice has a piece of quantum information $|\psi\rangle$ that she would like to send or "teleport" to Bob. Additionally, for this to work, Alice and Bob share an entangled Bell pair. Here are the consequent steps:

(1) First, Alice measures the two qubits A_1 and A_2 together in the *Bell basis*. Consequently, this breaks the entanglement between qubits A_2 and B, and causes qubits A_1 and A_2 to be entangled instead (this is monogamy of entanglement at play!).

(2) From this two-qubit measurement, Alice can extract classical information in the for of two classical bits, which she sends to Bob in a classical manner
(3) Bob, using the classical information, applies a correction on his quantum state, transforming it into the state $|\psi\rangle$ that Alice intended to teleport to Bob!

Key idea. Teleportation works by rewriting the three-qubit state in the Bell basis on Alice's two qubits! Then, by perfomring a projective measurement onto Alice's two qubits in the Bell basis, we will be left with a quantum state that is exactly the state that Alice had in qubit A_1 or a variation of that state, which is fairly simple to fix (correction unitaries).

Important features.

- No quantum information travels faster than light: Bob cannot recover $|\psi\rangle$ until he receives the two classical bits from Alice.
- The original state $|\psi\rangle$ is destroyed on Alice's side (because of measurement), preventing cloning.
- The protocol works for any dimension d by replacing Bell states with maximally entangled states in $\mathbb{C}^d \otimes \mathbb{C}^d$.

Teleportation is the foundational primitive behind quantum repeaters, gate teleportation, measurement-based quantum computation, and many quantum communication protocols.

5. (Ryan) Quantum Algorithms

5.1. A Classical Complexity Primer.

Why do people care about quantum computing? There are many practical problems where it is suspected that quantum computers will outperform classical computers. For example,

- Simulation of quantum systems (including quantum chemistry). This was Feynman's original motivation for introducing quantum computing.
- Optimization problems. This is the most exciting application because

  "Every problem in the world is an optimization problem!"

  (Prof. Lek-Heng Lim, STAT 28000)

  This is what has a lot of companies interested in quantum computing because optimization makes money. This also includes quantum machine learning.

But how would we actually *prove* that quantum computers can outperform classical computers? To do this, we need to introduce a notion of what makes one algorithm "faster" than another. The most common notion for this is *asymptotic time complexity*, or informally *big O notation*.

Let's imagine we're searching for an item in a list of length n. The most basic algorithm for this is *linear search*, where we check the first item, then the second, and so on until we find the item we want.

Definition 5.1. (not quite a definition) Assume the leading order term of the runtime $r(n)$ is some multiple of $f(n)$. Then the *asymptotic time complexity* is denoted $O(f(n))$.

For linear search, on average we have to search $\frac{n}{2}$ items, so the asymptotic time complexity is $O(n)$. This matches our intuition that if the list is twice as long, the algorithm should take around twice as long. It doesn't matter if the comparisons take 5 nanoseconds or 5 years. As long as they take a constant time, the time complexity is $O(n)$.

If our list is *sorted*, however, we can do much better. We can employ *binary search*, where we first check the midpoint of the list. If the midpoint is less than our item, we throw away the half of the list less than the midpoint, and if the midpoint is greater, we throw away the half of the list greater than the midpoint. We repeat until we find the item. This has an asymptotic time complexity of $O(\log(n))$. If the length of the list doubles, we add a constant amount to our average runtime.

For sufficiently large n, an $O(\log(n))$ algorithm will always be faster than $O(n)$. This is how computer scientists are able to determine whether one algorithm is faster than another. Be careful though: this is only the *asymptotic* time complexity and says nothing about the "small" n regime. There are algorithms which are technically the "fastest" but only become faster when n is larger than the number of atoms in the universe! (These are called *galactic algorithms*.)

5.2. Quantum Advantage.

Now that we know what makes one algorithm faster than another, how many *useful algorithms* with *provable quantum speedups* are there? Currently, only 2.

The one that makes headlines is *Shor's algorithm*, an algorithm that factors numbers into primes quickly. This has applications to cryptography, which is the source of all of the "quantum computing breaks encryption" claims (even though quantum-proof encryption is being implemented faster than actual quantum computing). This algorithm really deserves its own lecture, but the reason it's so much faster is that it relies on the *quantum Fourier transform*.

The *discrete Fourier transform* is a classical algorithm for signal processing (among many other things) that traditionally is $O(n^2)$. In the early days of computing, this made it infeasible to do signal processing in real-time. This was until the *fast Fourier transform* was discovered (or rather rediscovered by computer scientists) in the 1960s with a runtime of $O(n \log(n))$, which suddenly made signal processing (and many of its applications) feasible. The *quantum Fourier transform* has the potential to do it all again with a runtime of $O(\log(n)^2)$, and first among the improvements is Shor's algorithm.

The other one is *Grover's search*, an algorithm for finding an item in an unsorted list. Instead of running in $O(n)$ like linear search, it runs in $O(\sqrt{n})$. The reason this is useful is because there are many problems (called *NP* problems) where a solution is easy to verify but hard to find (e.g. solving a Sudoku). We can solve any of these using brute-force search, reducing the problem to finding an item in an unsorted list.

5.3. **Grover's Search Algorithm.** In order to run Grover's search, we need to first construct an *oracle*, which is a fancy CS-theoretic way of saying the easy verification circuit we're assuming we have. The oracle $\hat{O}$ acts on the computational basis as follows

$$\hat{O} |b_1 \dots b_n\rangle = \begin{cases} -|b_1 \dots b_n\rangle & \text{if the basis state is the solution} \\ |b_1 \dots b_n\rangle & \text{if it's not the solution} \end{cases}$$

We're not assuming we know the solution beforehand. We're only assuming that if we were handed the solution, we would be able to verify it. Just like every other treatment of Grover's search, I'm not going to tell you how to construct this oracle, since it depends on the exact problem, but if we can construct the oracle classically, we can construct it out of quantum gates. We'll assume this oracle runs in constant time, which is a little unrealistic but all that really matters is that it depends on n sufficiently weakly.

Once we have this oracle, you might think we can do even better than $O(\sqrt{n})$. Here's something that looks like a constant-time quantum algorithm.

(1) Initialize a uniform superposition of all possible qubit states

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{\text{bitstrings}} |b_1 \dots b_n\rangle$$

(2) Apply the oracle $\hat{O} |s\rangle$
(3) Figure out which bitstring flipped

If quantum computing really was just parallel computing over superpositions, this would be totally valid. Unfortunately, quantum computing doesn't work like that. Step 3 isn't possible to do in one step without already knowing the bitstring, so we have to be a bit more clever.

Note that our "oracle" is really just a reflection along the plane perpendicular to our solution state $|\omega\rangle$

$$\hat{O} = U_\omega = I - 2 |\omega\rangle \langle\omega|$$

Likewise, we can define a reflection along the plane that $|s\rangle$ is in.

$$U_s = 2\,|s\rangle\,\langle s| - I$$

If we define

$$|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{\text{bitstrings} \neq \omega} |b_1 \dots b_n\rangle$$

then observe that any state in the span of $\{|s'\rangle\,, |\omega\rangle\}$ remains in the span of $\{|s'\rangle\,, |\omega\rangle\}$, so we can visualize the whole thing on a 2D plane.

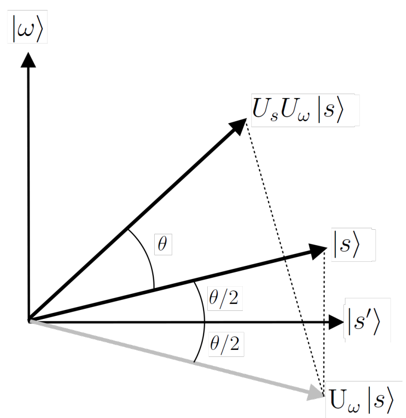


FIGURE 2. Grover's algorithm visualization (from Wikipedia)

The composition of two reflections is a rotation. Since the angle between $|s\rangle$ and $|s'\rangle$ is $\arcsin\left(\frac{1}{\sqrt{N}}\right)$, the rotation is by an angle of

$$\theta = 2\arcsin\left(\frac{1}{\sqrt{N}}\right) \approx \frac{2}{\sqrt{N}}$$

for large N. If we iterate U_sU_ω k times, the angle becomes $\left(k+\frac{1}{2}\right)\frac{2}{\sqrt{N}}$. The probability of measuring $|\omega\rangle$ is maximized when θ is closest to $\frac{\pi}{2}$, which happens when $k \approx \frac{\pi}{4}\sqrt{N}$. Grover's search can therefore be summarized as follows

(1) Initialize $|s\rangle$
(2) Apply U_sU_ω until the angle is closest to $\frac{\pi}{2}$. This will be $\lfloor \frac{\pi}{4}\sqrt{N} \rfloor$ times.
(3) Measure the outcome

This is transparently $O(\sqrt{N})$, a significant improvement from $O(N)$.

6. (Alice) Quantum Error Correction

6.1. **Probability density matrices.** Before talking about quantum error correction, we'll take a brief detour in the crucial topic of density matrices. As we've discussed, a pure quantum state can be written as the sum

$$|\psi\rangle = \sum_i \alpha_i |\psi_i\rangle ,$$

where $\{|\psi_i\rangle\}$ is a basis of energy eigenstates and

$$\sum_i |\alpha_i|^2 = 1.$$

We interpret each α_i as the probability of finding $|\psi\rangle$ in the state $|\psi_i\rangle$. But what about mixed states, which cannot be written in this way? We think of these as an *ensemble* of many quantum states. For these, it is necessary to define the *probability density operator*

$$\hat{\rho} = \sum_{m,n} \rho_{mn} |\psi_m\rangle \langle\psi_n| , \tag{6.1}$$

with the properties

$$\hat{\rho}^\dagger = \hat{\rho}, \mathrm{tr}\hat{\rho} = 1. \tag{6.2}$$

We can interpret the diagonal matrix elements ρ_{jj} as the "classical" probabilities of finding a state in the ensemble as $|\psi_j\rangle$. The off-diagonal matrix elements ρ_{ij} can be interpreted as the *coherence*, that is, they encode the degree of quantum interference between the states $|\psi_i\rangle$ and $|\psi_j\rangle$. Hence, $\rho_{ij} \neq 0$ if the ensemble contains a superposition of $|\psi_i\rangle$ and $|\psi_j\rangle$. Furthermore, as in Lecture 2, each ρ_{ij} contains a global phase and magnitude.

Observe that $\hat{\rho}$ represents a pure state (i.e. an ensemble with all states the same) if and only if the following hold:

- there exists some $|\Psi\rangle$ such that $\hat{\rho} = |\Psi\rangle\langle\Psi|$ (this is the state of all parts of the ensemble),
- $\hat{\rho} = \hat{\rho}^2$ and so $\mathrm{tr}\hat{\rho}^2 = 1$.

6.1.1. *Examples.* Consider a simple two-state system with basis $\{|0\rangle, |1\rangle\}$. Then the following ensembles have the corresponding density matrices:

- For an ensemble with half of the states $|0\rangle$ and the other half $|1\rangle$,
$$\hat{\rho} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$
- For an ensemble with all of the states in the superposition $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$,
$$\hat{\rho} = |\psi\rangle\langle\psi| = \frac{1}{2}\begin{pmatrix} 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

6.2. **The stabilizer formalism.** Now that we've defined the probability density matrix, what is quantum error correction exactly? The fundamental problem of quantum computing is that the pure states and entanglement that we exploit will *decohere*, i.e. collapse into noisy mixed states. This is the error that we are correcting against. The *stabilizer formalism* gives us a theoretical framework for thinking about this.

Recall the Pauli matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Note that each Pauli matrix has eigenvalues in $\{\pm 1\}$, and any two Pauli matrices either commute or anticommute. Furthermore, these generate the *Pauli group*[3] G of all operators we can perform on a single qubit. Thus, we can define

$$\Pi := \{I, X, Y, Z\}, \tag{6.3}$$

and we can write

$$G = \{e^{i\phi}\sigma \| \ \phi \in \{0, \pi/2, \pi, 3\pi/2\}, \sigma \in \Pi\}. \tag{6.4}$$

Then for a system of n qubits, we have as a group of possible operators

$$\Pi^n := \{e^{i\phi}\sigma_1 \otimes \cdots \otimes \sigma_n \| \ \phi \in \{0, \pi/2, \pi, 3\pi/2\}, \forall 1 \leq i \leq n, \sigma_i \in \Pi\}. \tag{6.5}$$

Note that these are *physical qubits* and not necessarily *logical qubits*, which are what we actually use for computation and must be tolerant against decoherence. For these, we define the *stabilizer subgroup* $\mathcal{S}$ as an Abelian subgroup of Π^n; in other words, elements within $\mathcal{S}$ commute. Then the *code-base*, which is where we have the logical qubits, is the eigenspace of operators in $\mathcal{S}$ for the eigenvalue $+1$. If the code-base has dimension 2^k (i.e. there are k logical qubits), then we call this the $[n, k]$ *stabilizer quantum error-correcting code*.

But why is the stabilizer subgroup "error-correcting"? Observe that from the Pauli matrices, a decohering error $E \in \Pi^n$ either commutes or anticommutes with any $g \in \Pi^n$. This gives us two possible prescriptions:

- If there exists a $g \in \mathcal{S}$ such that E and g anticommute (i.e. its *syndrome* contains -1), E is correctable.
- If E commutes with $\mathcal{S}$, it is correctable if and only if $E \in \mathcal{S}$.

If you know group theory, the neat and tidy summary is that any errors $E_1, E_2 \in \Pi^n$ can be corrected if $E_1^\dagger E_2 \notin \mathcal{Z}(\mathcal{S})$ or $E_1^\dagger E_2 \in \mathcal{S}$.

6.3. **The toric code.** Alexei Kitaev's *toric code* is one of the most important applications of the stabilizer formalism, since it introduced the field of topological quantum error correction, which we all know is well-loved by the board of UC Quantum. This is the most simple model out there, but still involves some annoying math (grr alg top[4]), so we provide only an introduction here.

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[3]If you don't know what this is exactly, just think of a *group* as a set closed under multiplication. Likewise, a subgroup is a closed subset.

[4]The lion does not concern himself with Peter May's comments on his REU paper.

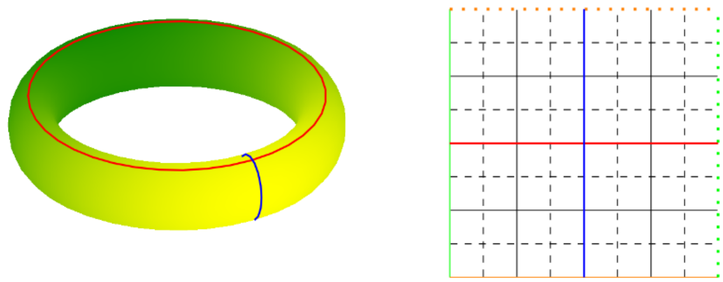

FIGURE 3. Mapping from the torus to the 2D hybrid lattice (stolen from my REU paper from 2 years ago).

The toric code is based on the fact that topologically, a torus is the same as a 2-dimensional $k \times k$ lattice (see Figure 3). After creating this original lattice (solid lines in Figure 3), it is useful to add the *dual lattice* (dotted lines in Figure 3) in order to form a combined *hybrid lattice*. Each unit length on the primal lattice represents a physical qubit, so any operator in $\Pi^n =: \text{TOR}(k)$ is just an assignment of some $\sigma \in \Pi$ to each such unit length.

Without going into gratuitous homological detail, this construction induces some really interesting behavior. The stabilizer operators are *vertex* and *plaquette* operators, while the Pauli operators on the 2 logical qubits correspond to noncontractible loops on the lattice. Resultingly, $\text{TOR}(k)$ can detect $k-1$ errors and correct for $\lfloor \frac{k-1}{2} \rfloor$.

In particular, errors are corrected using noncommutative elements, which correspond to chains on either the primal or dual lattice intersecting with contractible loops on the other. These can be thought of physically as moving quasiparticles (i.e. energy excitations) of different types around each other. But what does this really mean?

6.3.1. *Anyons!* The underlying physics of the toric code is really exciting and tied to many broader subfields. From our construction, we have that exchanging the states of these two types of quasiparticles introduces a factor of -1. This is very different behavior from our regular schmegular fermions or bosons, and so we have a new type of quasiparticle- *anyons*!

Anyons have been observed in some 2-dimensional systems, but never in 3 dimensions. Since they are not fermions but cannot occupy the same state, they are exceptionally stable, and so "braiding anyons" has become the basis for the modern race for a topological quantum computer. In particular, the elusive *Majorana particle*, which unlike Microsoft would have you believe has never been found, comes from braid defects in the toric code.